



Vidyavardhini's College of Engineering & Technology

Department of Civil Engineering

Course Name: Building Materials and Concrete Technology (2093113) **Sem/Year:** III/SE
Academic Year: 2026-27

Assignment No.: 06

Date of Submission: 19/10/2026

Marks: 10

2093113.6: Students are able to apply knowledge of special concreting and non-destructive testing based on site conditions and code requirements.

Q.No	Questions	BL
Q.1)	A reinforced concrete building is suspected to have reduced concrete strength. Select a suitable Non-Destructive Testing (NDT) method to assess the quality of concrete, interpret the pulse velocity rating obtained, and state the limitations of the selected NDT method.	3
Q.2)	A Ready-Mix Concrete (RMC) plant is to be established for a large construction project on a site measuring 60 m × 40 m. Identify the major components of the RMC plant and prepare a labelled layout showing the material flow during concrete production, ensuring efficient utilization of the available area.	3
Q.3)	A batch of Ready-Mix Concrete (RMC) is delayed in transit by 90 minutes during hot weather. Identify the problems that may occur in the concrete and recommend suitable corrective measures before placing the concrete.	3
Q.4)	A 10-year-old reinforced concrete slab shows surface cracks. Demonstrate the procedure for assessing its compressive strength using the rebound hammer test, explain how the test results are interpreted, and describe how the readings are correlated with core test results.	3
Q.5)	A contractor has to cast an M25 concrete slab at a daytime temperature of 40°C and another M25 slab at a night time temperature of 2°C. Prepare a concreting plan for both conditions by selecting suitable measures for mix design, placement, and curing, and justify how these measures will maintain concrete quality.	3

Signature of Subject Incharge